

```
$ export CFLAGS="-O3 -march=native -mtune=native"
```

Configuration

```
$ ./configure --enable-mods-shared="all ssl cache proxy auth_
alias mem_cache file_cache charset_lite dav_lock disk_cache" -
enable-nonportable-atomics=yes
```

Here, we run the *configure* script that's distributed with the source tree in the tarball, to configure the source tree. The `--enable-mods-shared=...` parameter is to compile the listed modules as Dynamic Shared Objects (DSOs), so that they can be enabled and disabled at will, after installation, by simply adding/removing a line in the configuration file.

The `--enable-mods-shared=all` doesn't actually do this for all modules; some are left out, and I have named those here.

In the above command, I have omitted two modules, *ldap* and *authnz_ldap*, which enable LDAP-based authentication. You should add the `--with-ldap` flag to the command in case you add these to the module list.

Once the *configure* script has run successfully (after eliminating any errors), proceed to compilation.

Compilation

```
make
```

If you have a multi-core processor—a dual-core or more—you can speed up the compilation by running it in parallel; instead of just running *make*, run *make -jN*, where N is the number of cores you have.

Installation

The last step, installation (copying the compiled binaries to their respective locations), is again made easy by the *make* tool:

```
$ sudo make install
```

...or,

```
$ su -c 'make install'
```

Installing files to their default locations, that is, */usr/local/apache2*, will require root permissions. If you have *sudo* (installed on every Ubuntu machine) you can use that, or if not, you will have to use *su -c*, in which case you need to know the *root* password.

You can issue the above command without *sudo* or *su -c* in case you supplied a path to the *configure* script (`--prefix=PATH`) that is writeable by your account.

So, now that you have installed Apache, let's move on to tweaking its configuration.

Apache configuration files

Apache uses a specific configuration file syntax, commonly found in many Linux applications.

The syntax includes certain directives, like *EnableSendFile On*, and some sections that resemble a kind of mark-up language, like what follows:

```
<Directory /path/to/directory>
```

```
# Directory configuration
```

```
</Directory>
```

The # denotes a comment, like in shell scripts.



Note: Assuming you have done the default configuration, i.e., with `--prefix=/usr/local/apache2` (which is the default, if you don't specify this option), Apache places your configuration files in */usr/local/apache2/conf*. If you passed any custom path prefix to the *configure* script, or a custom location for *etcd*, then your configuration files will be in the *conf* sub-directory of your custom prefix path, or in the custom *etcd* location you've given. We will assume the default location.

Apart from the main configuration files in */usr/local/apache2/conf*, it is possible to have directory-specific configuration placed inside the directories themselves. These do not contain a large amount of configuration, but are quite useful in case you want to change one single setting for one directory, when it makes no sense to add a `<Directory>` section to the main configuration file and restart/reload Apache. These files are named *.htaccess*, and it is possible to allow/disallow usage of one such file.

Here, I'll discuss only the main aspects of the configuration file—the important performance- and security-related options, along with certain basic ones. For all other options, you can find comprehensive documentation at <http://apache.org/docs/>

Directives, their usage and effects

Listen

This directive tells Apache which ports and IP addresses to listen on for requests. Many people miss this during configuration of virtual hosts, and particularly SSL hosts. Unnecessarily adding Listen directives will cause Apache to listen on those ports, causing a security threat.

The syntax is:

Listen [<ip-address>*]:*<port>**. For example:

```
Listen 80
```

```
Listen 11.22.33.44:80
```